Biochemistry Cume
January 16, 2004

One of the major problems in biochemistry is the separation of molecules of interest for analytical or preparative purposes from the complex mixtures found in organisms. Answer the following questions about biomolecular separations.

1. The technique known as affinity chromatography is commonly used in biochemistry. Explain how this technique works in general terms and then give a specific example of how one might use it to purify an enzyme that you are studying. 25 points

2. SDS-PAGE is used quite frequently as well to separate proteins by their molecular weight. Explain how this technique works. In particular, be sure to explain how proteins with very different isoelectric points can be reliably separated by molecular weight. 20 points

3. Size exclusion chromatography is also used to separate proteins by their molecular weight. Explain how this technique works. 20 points

4. SDS-PAGE is nearly always used analytically and size exclusion chromatography is usually used preparatively. Explain why. 10 points.

5. Differently sized pieces of DNA are often separated by agarose gel electrophoresis. Since the force exerted on a molecule in an electric field is dictated by the mass to charge ratio and the mass to charge ratio for any piece of DNA more than a few bases long is effectively constant, one would be lead to predict that DNA should move at a constant velocity regardless of size. What additional factor allows this technique to separate molecules on the basis of size? 20 points.